Dirac equation 3

Let

$$\psi_1 = \sqrt{E+mc^2}\begin{pmatrix} 1 \\ 0 \\ \frac{p_z c}{E+mc^2} \\ \frac{(p_x+ip_y)c}{E+mc^2} \end{pmatrix} e^{-i\xi/\hbar}$$

wavefunction for fermion spin up

$$\psi_2 = \sqrt{E+mc^2}\begin{pmatrix} 0 \\ 1 \\ \frac{(p_x-ip_y)c}{E+mc^2} \\ \frac{-p_z c}{E+mc^2} \end{pmatrix} e^{-i\xi/\hbar}$$

wavefunction for fermion spin down

$$\psi_3 = \sqrt{E+mc^2}\begin{pmatrix} \frac{p_z c}{E+mc^2} \\ \frac{(p_x+ip_y)c}{E+mc^2} \\ 1 \\ 0 \end{pmatrix} e^{i\xi/\hbar}$$

wavefunction for antifermion spin up

$$\psi_4 = \sqrt{E+mc^2}\begin{pmatrix} \frac{(p_x-ip_y)c}{E+mc^2} \\ \frac{-p_z c}{E+mc^2} \\ 0 \\ 1 \end{pmatrix} e^{i\xi/\hbar}$$

wavefunction for antifermion spin down

where

$$\xi = p_\mu x^\mu = Et - p_x x - p_y y - p_z z$$

and

$$E = \sqrt{p_x^2c^2 + p_y^2c^2 + p_z^2c^2 + m^2c^4}$$

Verify the wavefunctions solve the Dirac equation

$$i\hbar\left(\gamma^0\frac{\partial}{\partial t} + c\gamma^1\frac{\partial}{\partial x} + c\gamma^2\frac{\partial}{\partial y} + c\gamma^3\frac{\partial}{\partial z}\right)\psi = mc^2\psi$$

and are normalized as

$$|\psi|^2 = 2E$$

Verify dimensions.

$$\frac{Et}{\hbar} = \frac{[\mathrm{J}]\,[\mathrm{s}]}{[\mathrm{J\,s}]} = [1]$$

$$\frac{p_x x}{\hbar} = \frac{[\mathrm{kg\,m\,s^{-1}}]\,[\mathrm{m}]}{[\mathrm{J\,s}]} = [1]$$

$$\hbar\frac{\partial}{\partial t} = [\mathrm{J\,s}]\,[\mathrm{s^{-1}}] = [\mathrm{J}]$$

$$\hbar c\frac{\partial}{\partial x} = [\mathrm{J\,s}]\,[\mathrm{m\,s^{-1}}]\,[\mathrm{m^{-1}}] = [\mathrm{J}]$$

$$mc^2 = [\mathrm{J}]$$

Eigenmath script